

工科数学分析期末试题 (A 卷) 评分标准

一. 填空题 (每小题 4 分, 共 20 分)

$$1. \quad x - 2y - z + 2 = 0; \quad \left(\frac{3}{2}, 2, -\frac{1}{2}\right) \quad \dots\dots\dots 2 \text{ 分}, 2 \text{ 分}$$

$$2. \quad \frac{1}{2} \quad \dots\dots\dots 4 \text{ 分}$$

$$3. \quad \frac{2}{3}\pi a^4 \quad \dots\dots\dots 4 \text{ 分}$$

$$4. \quad \{0, 1, y-1\} \quad \dots\dots\dots 4 \text{ 分}$$

$$5. \quad \ln x = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} (x-1)^{n+1} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} (x-1)^n, \quad x \in (0, 2] \quad \dots\dots\dots 2 \text{ 分}, 2 \text{ 分}$$

二.

$$\text{解: } I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^a \rho^2 \sin^2 \theta \cdot \rho \cdot \rho d\rho \quad \dots\dots\dots 4 \text{ 分}$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 \theta d\theta \int_0^a \rho^4 d\rho$$

$$= 2 \cdot \frac{1}{2} \cdot \frac{\pi}{2} \cdot \frac{1}{5} a^5 \quad \dots\dots\dots 7 \text{ 分}$$

$$= \frac{\pi}{10} a^5 \quad \dots\dots\dots 8 \text{ 分}$$

三.

解: $f'_x(x, y) = 3x^2 - 6x = 0$, $f'_y(x, y) = 3y^2 - 6y = 0$,

得四个驻点 $(0, 0)$, $(0, 2)$, $(2, 0)$, $(2, 2)$ 3 分

$A = f''_{xx}(x, y) = 6x - 6$, $B = f''_{xy}(x, y) = 0$, $C = f''_{yy}(x, y) = 6y - 6$ 4 分

对 $(0, 0)$, $AC - B^2 = 36 > 0$, $A = -6 < 0$,

$(0, 0)$ 是极大值点, $f(0, 0) = 0$ 为极大值;5 分

对 $(0, 2)$, $AC - B^2 = -36 < 0$, $(0, 2)$ 不是极值点;6 分

对 $(2, 0)$, $AC - B^2 = -36 < 0$, $(2, 0)$ 不是极值点;7 分

对 $(2, 2)$, $AC - B^2 = 36 > 0$, $A = 6 > 0$,

$(2, 2)$ 是极小值点, $f(2, 2) = -8$ 为极小值。8 分

四.

解: (1) 抛物面 $z = 1 + x^2 + y^2$ 在点 $(1, 0, 2)$ 处的法向量为

$$\vec{n} = \{2x, 2y, -1\}|_{(1, 0, 2)} = \{2, 0, -1\}$$

则所求切平面方程为 $2(x - 1) - (z - 2) = 0$, 即 $z = 2x$ 2 分

(2) 所求立体在 xoy 平面的投影区域为 $D_{xy}: x^2 + y^2 \leq 2x$, 则

$$M = \iiint_{\Omega} \frac{1}{\sqrt{x^2 + y^2}} dx dy dz \quad \text{.....3 分}$$

$$= \iint_{D_{xy}} \frac{[(1 + x^2 + y^2) - 2x]}{\sqrt{x^2 + y^2}} dx dy \quad \text{.....4 分}$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{2\cos\theta} \frac{(1 + \rho^2) - 2\rho\cos\theta}{\rho} \cdot \rho d\rho$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{2\cos\theta} [(1 + \rho^2) - 2\rho\cos\theta] d\rho \quad \text{.....6 分}$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (2\cos\theta - \frac{4}{3}\cos^3\theta) d\theta$$

$$= 4 - \frac{4}{3} \cdot 2 \cdot \frac{2}{3} = \frac{20}{9} \quad \text{.....8 分}$$

五.

解: 因为 $\frac{du}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx} + \frac{\partial f}{\partial z} \cdot \frac{dz}{dx}$,2 分

方程 $e^{xy} - y = 0$ 两边关于 x 求导, 得 $e^{xy}(y + x \frac{dy}{dx}) - \frac{dy}{dx} = 0$,

解得 $\frac{dy}{dx} = \frac{ye^{xy}}{1 - xe^{xy}} = \frac{y^2}{1 - xy}$ 4 分

方程 $e^z - xz = 0$ 两边关于 x 求导, 得 $e^z \frac{dz}{dx} - (z + x \frac{dz}{dx}) = 0$,

解得 $\frac{dz}{dx} = \frac{z}{e^z - x} = \frac{z}{xz - x}$ 6 分

故 $\frac{du}{dx} = \frac{\partial f}{\partial x} + \frac{ye^{xy}}{1 - xy} \cdot \frac{\partial f}{\partial y} + \frac{z}{e^z - x} \cdot \frac{\partial f}{\partial z}$

或 $\frac{du}{dx} = \frac{\partial f}{\partial x} + \frac{y^2}{1 - xy} \cdot \frac{\partial f}{\partial y} + \frac{z}{xz - x} \cdot \frac{\partial f}{\partial z}$ 8 分

六.

解: $I = \iint_{\Sigma} \frac{xdydz + (z+1)^2 dxdy}{\sqrt{x^2 + y^2 + z^2}} = \iint_{\Sigma} xdydz + (z+1)^2 dxdy$ 1 分

用高斯公式, 加一个面 $\Sigma_1: z=0, x^2 + y^2 \leq 1$ 取下侧,2 分

$$I = \iint_{\Sigma} xdydz + (z+1)^2 dxdy$$
$$= [\iint_{\Sigma + \Sigma_1^-} + \iint_{\Sigma_1^+} xdydz + (z+1)^2 dxdy] \quad \dots\dots\dots 3 \text{ 分}$$

$$= -\iiint_{\Omega} (3+2z) dxdydz + \iint_{D_{xy}} dxdy \quad \dots\dots\dots 5 \text{ 分}$$

$$= -3 \iiint_{\Omega} dxdydz - 2 \iiint_{\Omega} z dxdydz + \pi$$

$$= -2\pi - 2 \int_{-1}^0 zdz \iint_{D_z} dxdy + \pi$$

$$= -\pi - 2\pi \int_{-1}^0 z(1-z^2) dz$$

$$= -\frac{\pi}{2} \quad \dots\dots\dots 8 \text{ 分}$$

七. 解1: L 的参数方程为:
$$\begin{cases} x = \cos \theta \\ y = \sin \theta \\ z = \cos \theta + \sin \theta \end{cases}, \quad \theta: 0 \rightarrow 2\pi, \quad \dots\dots\dots 3 \text{分}$$

$$\begin{aligned} I &= \oint_L xzdx + xdy + y^2dz \\ &= \int_0^{2\pi} [(\cos^2 \theta + \sin \theta \cos \theta)(-\sin \theta) + \cos^2 \theta + \sin^2 \theta(\cos \theta - \sin \theta)]d\theta \quad \dots\dots\dots 6 \text{分} \\ &= \int_0^{2\pi} (\cos^2 \theta - \sin \theta)d\theta \\ &= \pi \quad \dots\dots\dots 8 \text{分} \end{aligned}$$

解 2: 利用斯托克斯公式, 平面 $\Sigma: z = x + y$ 取上侧, $D_{xy}: x^2 + y^2 \leq 1$

$$I = \iint_{\Sigma^+} \begin{vmatrix} dydz & dzdx & dxdy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & x & y^2 \end{vmatrix} = \iint_{\Sigma^+} 2ydydz + xdzdx + dxdy \quad \dots\dots\dots 4 \text{分}$$

$$= \iint_{D_{xy}} [2y(-1) + x(-1) + 1]dxdy = \iint_{D_{xy}} (1 - x - 2y)dxdy \quad \dots\dots\dots 6 \text{分}$$

$$\stackrel{\text{对称性}}{=} \iint_{D_{xy}} dxdy = \pi \quad \dots\dots\dots 8 \text{分}$$

八. 解: $\lim_{n \rightarrow \infty} \frac{4(n+1)^2 + 4(n+1) + 3}{2(n+1) + 1} \bigg/ \frac{4n^2 + 4n + 3}{2n + 1} = 1, \quad R = 1$

$x = \pm 1$ 时, 级数发散, 故级数的收敛域为 $(-1, 1)$ 。 $\dots\dots\dots 2 \text{分}$

当 $x \neq 0$ 时,

$$S(x) = \sum_{n=0}^{\infty} \frac{4n^2 + 4n + 3}{2n + 1} x^{2n} = \sum_{n=0}^{\infty} (2n + 1)x^{2n} + 2 \sum_{n=0}^{\infty} \frac{1}{2n + 1} x^{2n} \quad \dots\dots\dots 3 \text{分}$$

$$= \sum_{n=0}^{\infty} (2n + 1)x^{2n} + \frac{2}{x} \sum_{n=0}^{\infty} \frac{1}{2n + 1} x^{2n+1} = \sum_{n=0}^{\infty} (x^{2n+1})' + \frac{2}{x} \sum_{n=0}^{\infty} \int_0^x x^{2n} dx$$

$$= \left(\sum_{n=0}^{\infty} x^{2n+1} \right)' + \frac{2}{x} \int_0^x \sum_{n=0}^{\infty} x^{2n} dx = \left(\frac{x}{1-x^2} \right)' + \frac{2}{x} \int_0^x \frac{1}{1-x^2} dx$$

$$= \frac{1+x^2}{(1-x^2)^2} + \frac{1}{x} \ln \frac{1+x}{1-x}, \quad \dots\dots\dots 7 \text{分}$$

当 $x = 0$ 时, $S(0) = 3$ $\dots\dots\dots 8 \text{分}$

九.

解: $b_n = 0, (n=1, 2, \dots)$ 1 分

$$a_0 = \frac{2}{\pi} \int_0^\pi (x-1)dx = \pi - 2, \quad \dots\dots\dots 2 \text{ 分}$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^\pi (x-1) \cos nx dx = \frac{2}{n\pi} \int_0^\pi (x-1) d \sin nx = \frac{2}{n\pi} [(x-1) \sin nx]_0^\pi - \int_0^\pi \sin nx dx \\ &= \frac{2}{n^2\pi} \cos nx \Big|_0^\pi = \frac{2}{n^2\pi} (\cos n\pi - 1) = \begin{cases} 0 & \text{当 } n \text{ 取偶数时} \\ -\frac{4}{n^2\pi} & \text{当 } n \text{ 取奇数时} \end{cases} \quad \dots\dots\dots 4 \text{ 分} \end{aligned}$$

$$\text{则 } x-1 = \frac{\pi}{2} - 1 - \frac{4}{\pi} \left(\cos x + \frac{1}{3^2} \cos 3x + \frac{1}{5^2} \cos 5x + \dots \right) \quad (0 \leq x \leq \pi) \quad \dots\dots\dots 6 \text{ 分}$$

$$\text{当 } x=0 \text{ 时, } 1 = \frac{\pi}{2} - 1 - \frac{4}{\pi} \left(1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right), \text{ 则 } \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8} \quad \dots\dots\dots 8 \text{ 分}$$

十.

解: 对 $\frac{\partial f(x,y)}{\partial x} = (2x+1)e^{2x-y}$ 两边关于 x 求积分, 得

$$\begin{aligned} f(x,y) &= \int (2x+1)e^{2x-y} dx = \frac{1}{2} \int (2x+1) d e^{2x-y} = \frac{1}{2} [(2x+1)e^{2x-y} - \int e^{2x-y} d(2x-y)] \\ &= \frac{1}{2} [(2x+1)e^{2x-y} - e^{2x-y}] + \varphi(y) = x e^{2x-y} + \varphi(y) \quad \dots\dots\dots 2 \text{ 分} \end{aligned}$$

$$\text{又 } f(0,y) = \varphi(y) = y, \text{ 得 } f(x,y) = x e^{2x-y} + y, \quad \dots\dots\dots 3 \text{ 分}$$

$$\text{则 } \frac{\partial f(x,y)}{\partial y} = -x e^{2x-y} + 1$$

$$I = \int_L \frac{\partial f(x,y)}{\partial x} dx + \frac{\partial f(x,y)}{\partial y} dy = \int_L (2x+1)e^{2x-y} dx + (-x e^{2x-y} + 1) dy \quad \dots\dots\dots 4 \text{ 分}$$

$$\text{设 } P = (2x+1)e^{2x-y}, \quad Q = -x e^{2x-y} + 1,$$

$$\frac{\partial P}{\partial y} = -(2x+1)e^{2x-y} = \frac{\partial Q}{\partial x}, \text{ 与路径无关} \quad \dots\dots\dots 5 \text{ 分}$$

选路径 $y=x, x:0 \rightarrow 1$, 则

$$I = \int_L (2x+1)e^{2x-y} dx + (-x e^{2x-y} + 1) dy = \int_0^1 [(2x+1)e^x + (-x e^x + 1)] dx \quad \dots\dots\dots 7 \text{ 分}$$

$$= \int_0^1 [(x+1)e^x + 1] dx = \int_0^1 (x+1) d(e^x) + 1$$

$$= (x+1)e^x \Big|_0^1 - e^x \Big|_0^1 + 1 = e + 1 \quad \dots\dots\dots 8 \text{ 分}$$

十一.

解：因 $f(x)$ 单增，所以级数为正项级数，且

$$\lim_{n \rightarrow \infty} \frac{f(a + \frac{1}{n^p}) - f(a)}{\frac{1}{n^p}} = f'(a) \quad \dots\dots\dots 2 \text{ 分}$$

(1) 若 $f'(a) \neq 0$ ，则此级数与 $\sum_{n=1}^{\infty} \frac{1}{n^p}$ 有相同的敛散性，故

$$\text{当 } \begin{cases} p > 1 \text{ 时, 收敛} \\ p \leq 1 \text{ 时, 发散} \end{cases}, \quad \dots\dots\dots 4 \text{ 分}$$

(2) 若 $f'(a) = 0$ ，由泰勒公式知

$$f(a + \frac{1}{n^p}) - f(a) = \frac{f''(a)}{2!} (\frac{1}{n^p})^2 + o((\frac{1}{n^p})^2) \sim \frac{f''(a)}{2!} \cdot \frac{1}{n^{2p}}, \quad \dots\dots\dots 6 \text{ 分}$$

故此时原级数与级数 $\sum_{n=1}^{\infty} \frac{1}{n^{2p}}$ 有相同的敛散性，故

当 $p > \frac{1}{2}$ 时， $\sum_{n=1}^{\infty} \frac{1}{n^{2p}}$ 收敛，故原级数收敛；

当 $0 < p \leq \frac{1}{2}$ 时， $\sum_{n=1}^{\infty} \frac{1}{n^{2p}}$ 发散，故原级数发散，
\dots\dots\dots 8 \text{ 分}

综上所述：

$$\text{当 } f'(a) \neq 0 \text{ 时, 有 } \begin{cases} p > 1 \text{ 时, 收敛} \\ p \leq 1 \text{ 时, 发散} \end{cases}; \quad \text{当 } f'(a) = 0 \text{ 时, 有 } \begin{cases} p > \frac{1}{2} \text{ 时, 收敛} \\ 0 < p \leq \frac{1}{2} \text{ 时, 发散} \end{cases}$$